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Biomimetics for early stage biofouling prevention: templates from insect cuticles

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A biomimetic antifouling material study was carried out utilising superhydrophobic cicada and dragonfly wings replicated with a polymer (epoxy resin). They were tested in a marine biofouling study for up to 1 week in addition to biofouling assays of protein, carbohydrate and DNA absorption. The materials were compared against a commercial antifouling paint and a polymeric smooth surface constituting a control sample. The replicated surfaces demonstrated superior antifouling properties in comparison to the control and similar efficiency in DNA (10% reduction), protein and carbohydrate adsorption (15%) to the commercial anti-fouling paint. As the fabricated surfaces have roughness at the nanometre scale it is probable that the low adsorption properties, at least in the early stages, may be related to air trapped at the surface. Interestingly the most disordered replicated surface (dragonfly wing replicate) demonstrated the lowest values of absorption.

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Introduction

Biofouling and biofilms exist as a seemingly impossible limiting factor to be overcome by almost all water-related industries.^{1–4} The economic cost of biofouling is challenging and includes expenses such as increased fuel cost, loss of efficiency in data collection, added energy for new sea energy production systems, drag and manoeuvrability issues, invasive species transfer and general over-maintenance procedures.

When a substrate is immersed into a non-sterile aquatic medium, it rapidly becomes colonised with a biofilm and the subsequent effects of biofouling – both are not fully understood, owing to the complexity in variation, seasonality and temperatures of the sea. There are generally 3 stages in the biofouling process: (i) the adsorption of organic and inorganic compounds and other biochemicals, (ii) colonisation of micro-organisms and primary fouling organisms on the surface of the substrate and (iii) the attachment, growth and proliferation of higher macro-fouling organisms – known as macrofoulers.⁵ It is the presence of fouling that causes the deleterious effects for anyone faced with tackling the problem. Generally, the initial stages of biofouling are considered to be stochastic, where the

microbial component of the biofilm matrix is held fast and rendered irreversible. The permanency of the biofouling issue occurs when exopolymeric substances (EPSs) are produced forming a protective matrix, ensuring that the adhered microorganisms can thrive, live and grow upon the surface of the material.^{5,6}

Solutions to the biofouling problem have been developed from as early as the 5th century using tar, pitch, wax, copper and lead.⁷ More sophisticated antifouling approaches were developed in the 19th century (containing copper and rosin) and in the 1950's the incorporation of organotin compounds vastly improved the antifouling performance for many marine industries particularly using the tributyltin (TBT) moiety.⁸ Unfortunately, TBT and the use of TBT in self-polishing paints has been far from ideal. The paints have had detrimental environmental effects on non-target organisms in the marine environment leading to imposex in molluscs and accumulation in mammals and fish, forcing an outright ban through the International Maritime Organisation (IMO). These actions have led to an urgency in research to find a non-toxic, novel, environmentally friendly antifouling strategy.^{1,3,9}

Given the obvious recurring problems of non-target effects and furthermore, bioaccumulation, new research directions have emerged as non-toxic control strategies, in particular the application of foul-release and drive towards biomimetics.¹⁰ The field of biomimetics is defined as the study of the structure and function of a biological system and process as a model or biologically inspired informed-design for sustainable engineering of a material, process or problem.

There are many relationships between the epibiont (settler) and basibiont (host). For the basibiont there are a number of

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reasons why settlement on the surface is disadvantageous.¹¹ Primary colonising organisms, such as bacteria can damage the surface, reduce reproductive output, reduce light efficiency and cause pH fluctuation resulting in impaired surface function and is therefore deemed non-symbiotic. For example, marine plants will be severely impacted by biofouling on photosynthesis and growth. For bivalves epibionts may negatively affect the shell condition, growth rate and overall survival. Epibionts also increase drag and combine to make extra mass on the underlying organism, restricting mobility and lowering speed – all negative implications when biofouling occurs in marine industries or marine structures.

Defence mechanisms that prevent or reduce the effects of biofouling have been previously characterised as: behavioural, mechanical, physical or chemical.¹² However, the reality of the matter is that many organisms use hybrid or multi-component antifouling strategies as described in Chapman *et al.* (2014). Furthermore, the use of hybrid antifouling strategies will enhance the thought process in the design of more effective solutions. Consequently, the design and manufacture of a surface that can successfully prevent or control microfouling will, by and large, be able to prevent the succession of macrofouling organisms which will only arise if the microfouling stages can be controlled.

The work presented in this paper uses a series of land-flying insects as templates for their potential to prevent the effects of biofouling in the initial stages with a particular focus on marine applications. The insects were chosen as it is already known that their wing structure exhibits a superhydrophobic tendency¹¹ which is often associated with a desirable surface energy in preventing microbial adhesion. As well, as highlighted in a number of previous studies, some insect structuring demonstrates a nanostructured surface which can control, and importantly minimise, the interaction of some solid particles *e.g.*, natural organic contaminants such as pollens as well as hydrocarbons and silica particles, *e.g.* ref. 12 and 13. A natural extension as suggested in such studies is investigating solid contacts of insect cuticles under aqueous conditions.¹¹

Two insect groups of which many exhibit superhydrophobic properties include cicadas and dragonflies.^{11,13} In particular, the wings of these groups often possess nano-pillar scale arrays on the epicuticle. The thin wing membrane of these insects encounters a wide range of environmental conditions and is often exposed to periods of contact with water. As well, insects related to these groups (for example damselflies which are closely related to dragonflies – both in the Order Odonata) undergo full submersion of wings in water; for several hours in some cases when laying eggs. Typically, the structuring found in cicada and dragonfly wings (and damselflies) is in the range of several hundred nanometres in relation to spacing and height with the actual projections less than 100 nm in diameter at the apex.^{11–14}

The aim of this work was to create artificial biomimetic surfaces replicated from dragonfly and cicada wing membranes native to Australia. The replicated surfaces were then taken through comprehensive antifouling assays previously developed by the group and compared to a market rival antifouling coating.

The replicated materials were compared to the smooth material control group and tested in a marine water environment.

Materials and methods

A series of material replication steps were performed in order to generate similar topographies to cicada and dragonfly wing nanostructures. The replicated materials were then exposed to a marine water body in Yeppoon, Queensland, Australia, for 7 days. The materials were subsequently analysed for biochemical responses to biofouling in its initial stages and characterised with a series of biochemical assays. The epoxy matrix selected for this study had previously demonstrated significant reductions in microbial adhesion (Kharchenko *et al.*, 2012 and Chapman 2014) and represented an excellent starting point for this work.^{1,15} Epoxy cross-linked materials already provide a stratifying ability and constitute one of the most promising approaches to develop improved antifouling coatings with improved mechanical and adhesion properties. It is these properties that have given us the impetus to select this material for our study.

Insect wing collection and preparation

Samples were selected based on previous work by Watson *et al.* Cicadas (*Psaltoda claripennis* and *Tamasa tristigma*) were collected from the greater Brisbane parkland areas of Queensland, Australia (typically on flora such as eucalypts). *Tamasa tristigma* were also collected from bushland surrounding the Ross River in Townsville, Queensland. Dragonflies (*Rhyothemis graphiptera*) were collected from bushland surrounding the Ross River in Townsville. Wing samples (forewings) were excised from the insect bodies using a sterile scalpel. Wing membrane surfaces were cleaned prior to experimentation with a stream of Milli-Q water. Fig. 1 shows examples of the 3 insect species used in this study.

Chemicals and reagents

A liquid polymer polydimethyl siloxane (PDMS) kit (Sylgard 182 elastomer) was purchased from Dow Corning, Australia. Protein standard chemicals [bovine serum albumin (BSA)], Lowry reagent powder, Folin–Ciocalteu Phenol, glutaraldehyde, D-(+)-glucose, and D-glucuronic acid were purchased from Sigma Aldrich, Australia, and used without further purification. Epoxy resin (Araldite CY212) was purchased from Agar Scientific, Australia. The antifouling paint used to benchmark the materials—Cruiser Uno International—was purchased in Australia.

Physical characterisation of the insect wing replicate samples

The samples had been previously placed in a 2.5% glutaraldehyde solution in seawater to fix and preserve any material on the surface. Surface analysis of adhered microorganisms and cells of the washed artificial dragonfly-wing samples was carried out using a Jeol JSM 6360 scanning electron microscope (SEM). Accelerating voltages of 5–15 keV with secondary electron (SE) detection was used, varying the working distances from 5 to 10 mm. Nine replicate samples were prepared. The artificial wing sample of dimensions 1 cm × 1 cm was placed onto a carbon

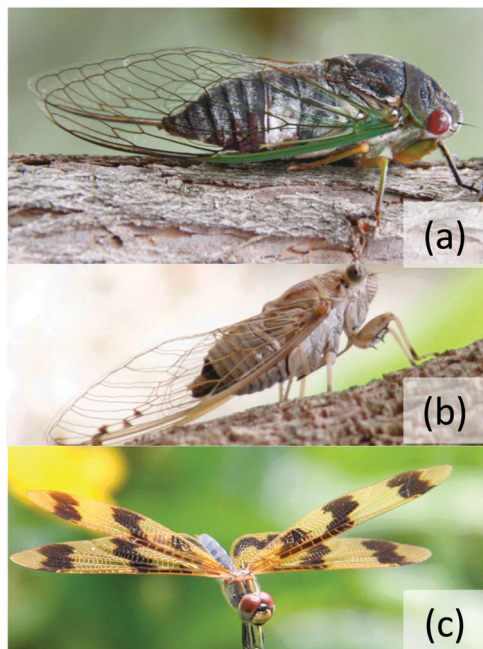


Fig. 1 Insect species used for replication in this study. (a) Sample 1 – *Psaltoda claripennis*, (b) sample 2 – *Tamasa tristigma* and (c) sample 3 – *Rhyothemis graphiptera*.

stub. The samples were sputter-coated with gold for 2 min, at which point an approximate deposition of a 25 nm gold layer was achieved. A comparison of the insect wing to the replicated sample was carried out to demonstrate the success of the replication process.

Attenuated total reflection-Fourier transform infra-red measurements of wings (ATR-FTIR)

Attenuated total reflection-Fourier transform infrared (ATR-FTIR) spectra of the wing sample membranes were recorded on a Perkin Elmer Spectrum Two FT-IR instrument, utilizing an ATR sampling technique, equipped with a diamond crystal.

The area of the sample to be measured was selected visually and pressure was added until contact was made between the surface of the wing membrane and the ATR crystal. Background spectra were obtained through the ATR element when it was not in contact with the sample. Reproducible contact was achieved by the use of an electronic sensor plate. Infrared spectra were collected at 1 cm^{-1} resolution utilising 16 scans in the range $3200\text{--}750\text{ cm}^{-1}$. Data were displayed after importing into an Excel package.

Contact angle measurements

Static contact angle measurements were determined using an optical microscope system equipped with a digital camera. Ten measurements per droplet (5 mL Milli-Q water) were taken on images captured under ambient conditions of $20\text{--}23\text{ }^{\circ}\text{C}$ and RH of $55\text{--}75\%$. Left and right angles between the sample surface and the tangent line to the droplet were considered as one measurement. Droplets were applied to each of the wing membranes, where possible, near the dorsal cell region between the wing veins (for cicada species). This region was used in order to accommodate the droplet footprint without the influence of the vein structure. The proximity of the vein structuring on the dragonfly necessitated angle measurements with the droplet in contact with vein structures. For illustrative purposes some measurements were taken with water dyed for contrast.

Artificial dragonfly topography replication

Negative replicas were produced by pouring PolyDiMethylSiloxane (PDMS) (Sylgard[®]-184) on whole wings held in a silicone rubber mould and polymerised at $85\text{ }^{\circ}\text{C}$ for two hours. PDMS was supplied by Dow Corning as a two-part silicone elastomer. The base and curing agent were mixed at a 10:1 weight ratio. After cooling, the wing tissue was pulled away from the PDMS cast leaving an impression that was used to produce a positive cast. The casts were then formed by the application of liquid Epoxy araldite resin, poured over the patterned PDMS mould and cured in an oven at $60\text{ }^{\circ}\text{C}$ for three days to fully cure the polymer, Fig. 2.

Marine deployment

In order to test the materials, the replicated insect wing samples were attached to glass slides in rows of three. Five glass slides containing a total of 15 samples per replicated insect wing samples were then deployed in engineered marine-grade racks suitable for marine water deployment, as previously described for a 7 day period, to assess the ability to withstand the effects of microfouling.¹⁶ The water was sampled from Yeppoon, Queensland, Australia; the temperature of the water was *ca.* $24\text{ }^{\circ}\text{C}$ during the month of July 2015.

Protein and carbohydrate assay

Protein adsorption was measured and carried out using a modified Lowry assay. Three reagents were prepared: (1) Lowry reagent powder was dissolved in 40 mL of Milli-Q water;

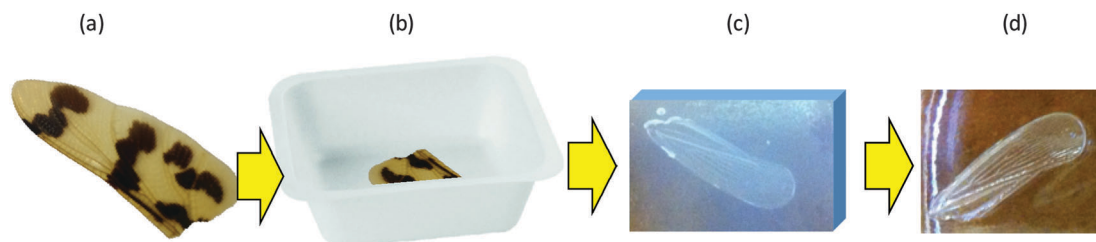


Fig. 2 Schematic representation of the replication stages of the natural wing surfaces in PDMS and epoxy resins. From sample preparation to replication this process takes 24 hours.

(2) Folin–Ciocalteu's phenol reagent was dissolved with a 1 : 5 ratio in Milli-Q water; and (3) BSA reagent was prepared by dissolving 2 mg of BSA in 5 mL of Milli-Q water to achieve a protein concentration of $400\ \mu\text{g mL}^{-1}$. A calibration curve was established. In a typical assay, Lowry reagent, Folin–Ciocalteu's reagent, and 0.5 mL of the scraped sample were mixed and then incubated for 30 min at room temperature. This solution was then analysed at 750 nm on a UV-visible spectrophotometer.

Carbohydrate adsorption was determined by taking a 0.5 mL aliquot of the sample. This was transferred into a 15 mL centrifuge tube, into which 0.5 mL of phenol solution and 2.5 mL of concentrated sulphuric acid were added immediately. The samples were then incubated for 10 min at room temperature and then at $30\ ^\circ\text{C}$ for 15 min. After a 5 min resting time, the samples were then measured at 480 and 490 nm using UV-visible spectrophotometry.

Deoxyribonucleic acid adsorption assay

In a typical assay DNA adsorption on the artificial dragonfly wing surfaces was determined using a commercial Quant-IT™ PicoGreen® dsRNA Reagent Kit (Invitrogen, Molecular Probes). A series of low and high calibration curves were derived using the following solution setup. Solution 1: $20\times$ TE buffer (200 mM Tris–HCl, 20 mM EDTA, pH 7.5) was diluted 20-fold in water for molecular biology. Solution 2: PicoGreen reagent was diluted 200-fold in $1\times$ TE buffer. Lambda-DNA stock solution was prepared as follows: a $2\ \mu\text{g mL}^{-1}$ (high range) stock solution was prepared by diluting the $100\ \mu\text{g mL}^{-1}$ Lambda-DNA standard 50-fold in $1\times$ TE buffer. A $50\ \text{ng mL}^{-1}$ (low range) stock solution was prepared by diluting the $100\ \mu\text{g mL}^{-1}$ Lambda-DNA standard 2000-fold in $1\times$ TE buffer. A series of low and high calibration curves were then derived from the calibration standards in the kit instructions. If necessary, samples were diluted in $1\times$ TE buffer to reach DNA concentrations within the range of one of the calibrations. One millilitre of standards or samples was transferred into a 1 cm disposable plastic cuvette, where 1 mL of solution 2 was added and incubated for 2 min in the dark at room temperature. The relative fluorescence of all the solutions was measured at the excitation wavelength of 480 nm and the emission wavelength of 520 nm.

Biomass and slime

Each replicated dragonfly wing substrate was weighed before and after immersion into Yeppoon Marina to ascertain mass fluctuation across the surface of each of the samples. Samples, following exposure, were rinsed with Milli-Q water to remove any unadhered material and then allowed to air-dry (until weight is constant) in order to eliminate water from the biomass result. A deduction of weight of each sample gave a result pertaining to the overall adhered content. On the same samples, slime (glycocalyx) was evaluated through a series of fixation and staining steps. Slime production was measured on the test material that was attached to the surface of the replicate dragonfly wing only. The samples, once washed, were reacted with Carnoy's solution (6 : 3 : 1, absolute ethanol : chloroform : glacial acetic acid) for 30 min. The solution was

then decanted and toluidine blue 0.1% (v/v) (Avonchem Ltd, UK) was added to stain the biofilm present on the polymers for 1 h. The excess stain was decanted off, and the polymers were rinsed twice with Milli-Q water (3 mL). NaOH (0.2 M) was added and the solution was heated to $85\ ^\circ\text{C}$ for 1 h. Each sample was then left to cool to room temperature and the optical density was measured at 590 nm (Cary 50 UV-visible spectrophotometer).

Statistical analysis

All experiments were performed in triplicate. Statistical analysis was performed using one-way analysis of variance (ANOVA) to evaluate the differences between groups. A p -value of $* < 0.05$ and $** < 0.01$ was considered statistically significant.

Results and discussion

Surface analysis

Following replication of the insect wings in epoxy resin a series of surface characterisation steps were performed to ascertain the physical and structural properties of the replicated materials compared to those of the natural surfaces. The surface morphology of the epoxy resin replicates closely resembles the physical characteristics of the natural surfaces as shown in Fig. 3.

The scanning electron micrographs of the cicada insect wing replicates revealed well-ordered arrays of conical projections with a similar spacing ($\sim 200\ \text{nm}$ for *Psaltoda claripennis* and *Tamasa tristigma*, samples 1 and 2). The height of the projections were comparable to the spacing of $\sim 200\ \text{nm}$. The (dragonfly – *Rhyothemis graphiptera*, sample 3) and its counterpart replica also consisted of nanopillars (average heights of $\sim 200\ \text{nm}$); however, they were more randomly arranged than the cicada samples (Fig. 3c). Thus, large areas covering 10 's of microns illustrate a homogenous type topography; however, at the individual scale of pillars the arrangement is somewhat random. Typically, in nature evanginations on the cuticle are not as rigid as we are demonstrating in this work. However, the materials used in our replication study do present as rigid superhydrophobic surfaces (a tensile modulus value of 15 GPa)¹⁷ in each of the surface characterisation steps. Upon comparing the insect wing a tensile modulus of 3.7 GPa is observed,¹⁸ demonstrating 5 times less rigidity in the natural surfaces. The heights of the surface pillars produce a superhydrophobic effect based on a Cassie–Baxter wetting state. Watson *et al.* have also reported similar heights for a natural wing surface topography of this type.¹¹ In general the copied surfaces are replicated with excellent reproducibility for a robust analysis of the biofouling properties, if any, on the developed materials.

All three natural surfaces demonstrated a superhydrophobic interaction with water. The static contact angles for the two cicada wings were within the range of 149 – 162° . The contact angle for the dragonfly wing was between 155 and 164° . For comparison, the epoxy without modification displays a contact water angle of 101° .¹⁹ These high values can be explained by the rough topography in combination with chemistry.

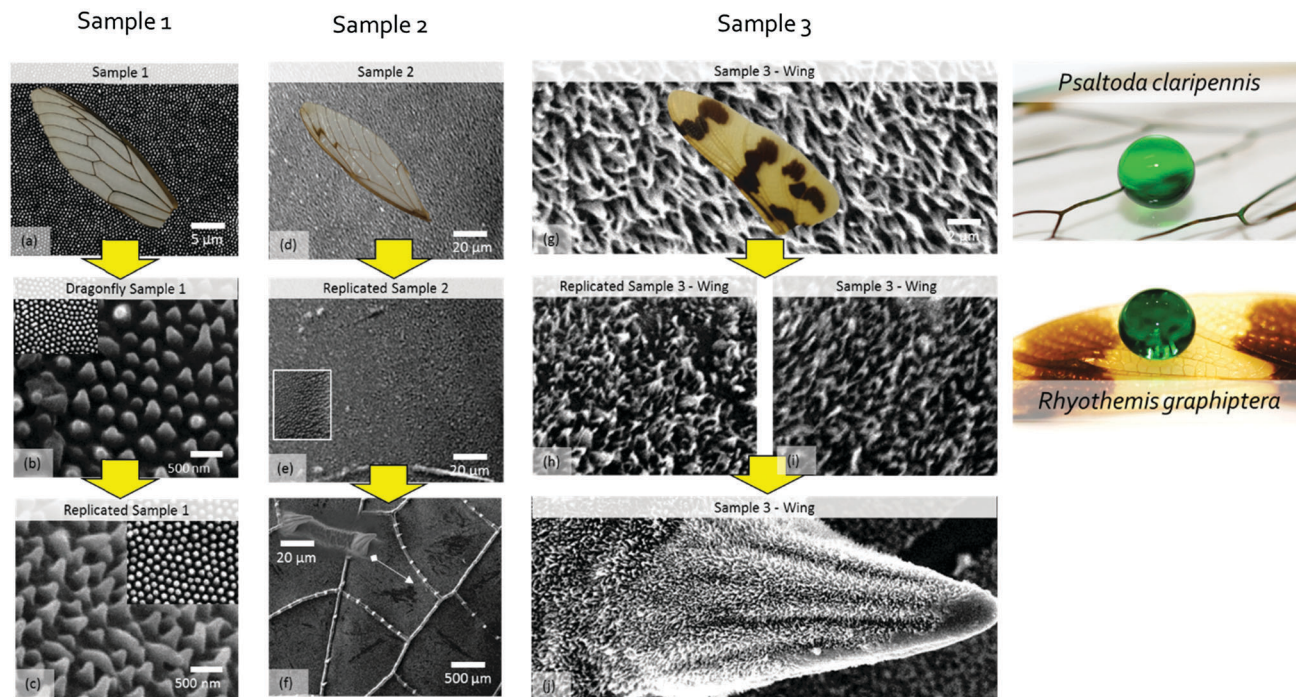


Fig. 3 Scanning electron micrograph comparisons of replicated wings and natural surfaces. (a) Natural surface of the wing of *Psaltoda claripennis*, (b) 1 μm natural sample and (c) replicated sample structure of sample 1 with 1 μm inset. (d) Natural wing structure of *Tamasia tristigma* (e) and (f) comparison of replicated epoxy wing and natural wing structure, (g) details of the natural wing structure illustrating the high surface features, (h) natural wing structure of sample 3, *Rhyothemis graphiptera*, (i) replicated structure and inset of surface features of the replicated epoxy sample and (j) showing expanded details of the replicated epoxy structure. Far left images showing the superhydrophobic properties of the wing surfaces.

Generally, insects as well as plants also have superhydrophobic compounds secreted on their surfaces. Notably, amides from protein, chitin and fatty acids all produce surface water repelling chemistry.²⁰

As shown in Fig. 4, attenuated transmission reflectance was used to characterise the surface chemistry of the natural samples. The spectra indicate the presence of lipids, along with other major structural components of insect wings, *i.e.*, proteins and chitin. Aliphatic hydrocarbons and esters were present which are attributable to epicuticular lipids. Bands corresponding to amide ($1500\text{--}1700\text{ cm}^{-1}$), ester ($1720\text{--}1750\text{ cm}^{-1}$), and alkyl chain ($2840\text{--}3000\text{ cm}^{-1}$) functional groups can be seen in representative spectra as shown in Fig. 4, with C-H₂ stretching (long-chain hydrocarbon) ester peaks attributed to waxes. The wax type components of the insect cuticle (and the corresponding low surface energy) may contribute

to the high efficiency of the replication of the surfaces as shown in Fig. 3.

Biofouling assays

A series of assays were used to assess the antifouling performance of the materials and the commercial antifouling paint. These assays are used to assess the biofouling levels and thus the overall antifouling performance of the materials following a 7 day study. The window of time has been carefully selected to represent the microfouling stage of the biofouling process. Numerous authors present data on the microfouling stage as this is the stage that, if prevented, can minimise the subsequent biofouling macrofouling stages. These marine water based tests quantify certain biochemical signatures and components of biofilm and the overall biofouling process characterised by the material adhered to the test materials.

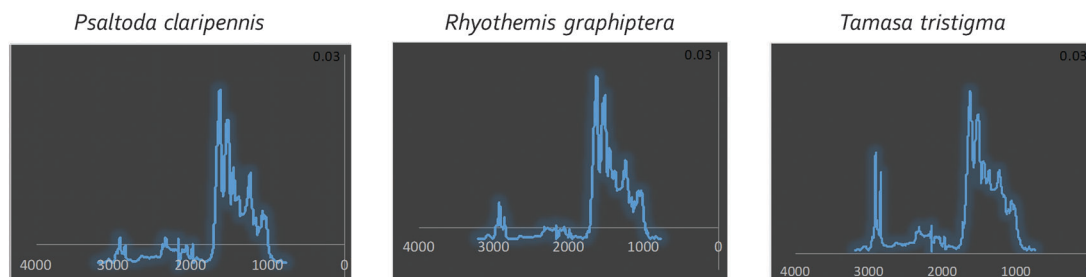


Fig. 4 ATR-FTIR measurements of the surface chemistry of the natural samples. x-Axis measured in wavenumber cm^{-1} and y-axis measured in intensity.

Protein adsorption

Protein adsorption is an important process in determining the biochemical composition of the antifouling material. Protein is a direct indication of biofouling generated through conditioning and furthermore the production of proteinaceous compounds which are naturally occurring and also produced by microbes and macrofouling organisms.

Following a comprehensive 1 week study, replicated insect wing samples were assessed for protein adsorption, Fig. 5. Many authors use the level of protein adsorbed on a surface as a measurement tool to predict fouling.²¹ Furthermore, it is widely recognised that carbohydrate and protein are the main components of EPSs; the debate is rife on whether carbohydrates or proteins influence subsequent stages of biofouling but we consider them both of vital importance. The antifouling paint was significantly effective in this assay. The one-way ANOVA for the 1-w data for each sample concludes that there was a statistical difference between the mean values of the groups determined by one way ANOVA $F = 37.61$ $p < 0.001$. Furthermore, following Tukey's *post hoc* test, there was a significant difference between the insect wings, antifouling paint and the blank epoxy sample ($p < 0.01$). The temperature of the fouling environment contributes greatly to the increased metabolism of microorganisms and justifies why the biochemical levels are higher than those reported in cooler water environments.²² All the replicated samples showed a reduced level of protein adsorption compared to the smooth control surface (Fig. 4) and compared favourably with the commercial paint (within 10% of the commercial product). Sample 3 demonstrated the highest resistance to protein adsorption when compared to the other samples for longer time periods. A 50% reduction of fouling was observed in sample 3 when compared to the epoxy smooth blank. This suggests that the more disordered nanostructure represented by the dragonfly structure is more effective in reducing protein adhesion than the more ordered cicada replicated surfaces for periods over several days or longer. A more disordered topographical

structuring has also been observed recently to be more effective in inhibiting the adhesion of bacteria (both Gram negative and positive) where a stretching mechanism has been suggested resulting in the disruption of the surfaces in contact with the nanostructuring.^{12,14,23,24}

Carbohydrate assay

Carbohydrates are also used as an indicator for microbial activity on a given surface.²⁵ In particular, the level of carbohydrates attached to a substrate is indicative of the biofilm developed through microbial activity.²⁶

Fig. 6 shows the total carbohydrate adsorption values for the insect replicated materials compared to the antifouling paint and the epoxy blank samples. All materials on the first day had very similar carbohydrate adsorption levels on their surfaces. In the extended microfouling stage at 1 week, sample 3 and the antifouling paint have demonstrated least carbohydrate adsorption. The results for carbohydrate adsorption for sample 3 reflect a similar result to the protein adsorption in Fig. 5. Indeed, in this case, the replicated surface demonstrated a superior resistance in the early stages and of equal performance after 1 week in comparison to the paint product. Thus, just like protein absorption, the disordered nanostructured surface appears to be resistant to carbohydrate accumulation.

DNA adsorption

Another key biochemical present in a biofilm network is in the form of DNA which is readily produced in biofouling organisms and also attributed to organism interaction, shown in Fig. 6. The use of DNA to characterize biofouling levels on surfaces is reported frequently.^{6,25}

The level of DNA on the surface of the substrate is directly linked to microbial and biofilm-forming species and can be quantified using the commercially available quantification method developed by Black *et al.* in 2004.²⁷ In a recent publication by Chapman *et al.* DNA adsorption was reported as a characterisation

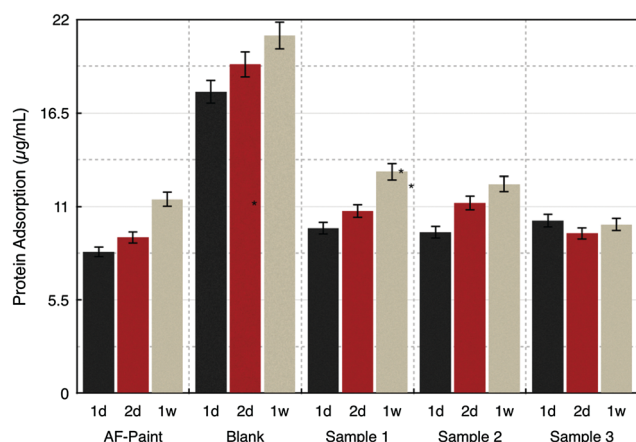


Fig. 5 Bar plot showing protein adsorption observed on materials following a 7 day marine biofouling study. Sample 3 and the AF paint have demonstrated a 50% lower protein adsorption value when compared to the epoxy blank. ($n = 14$, +1 SD).

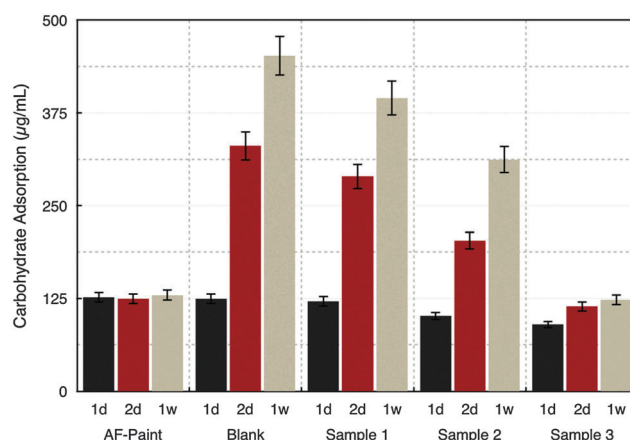


Fig. 6 Bar plot showing the overall carbohydrate adhesion following a 7 day marine biofouling study. Sample 3 has demonstrated excellent protein and carbohydrate adhesion during the marine study ($n = 14$, +1 SD).

method for biofouling characterisation at the microfouling stages of the overall fouling process.¹

Similarly, some authors have shown that biological molecules such as DNA or RNA have an ability to assemble on smooth surfaces rather than surfaces that have architectural features such as pillars and troughs.²⁸ Our data support this finding with all roughened samples exhibiting low adsorption levels in comparison with the control smooth polymeric surface (epoxy blank). All three insect replicates exhibited a greater resistance to DNA adsorption after a period of one day. The disordered dragonfly replicated surface demonstrated the lowest adsorption of all the surfaces for all time periods clearly outperforming the commercial product and in accord with our other absorption experiments.

Biomass and slime

A final biochemical assay that was used in determining the anti-fouling performance of the substrates was run in tandem

to all other assays used in this study. Slime production and the overall mass change (% weight increase) are reported in Fig. 8. Slime, given by the level of glycocalyx (inclusive of excreted EPSs) adsorbed on each of the materials and infers microbial proliferation and interaction of the surface in a microfouling capacity. Glycocalyx, in this biofouling assay, measures the enriched coating surrounding the eukaryotic and prokaryotic cells, particularly focusing on bacteria. In another biofouling group's work, slime analysis has proved to be an effective assay in determining the overall biofouling activity; however, only when run in conjunction with concrete quantitative biochemical assays.²⁵ The 'slime test' can be used as a confirmatory screening assay, as demonstrated in our previous group's work.²⁹ The best performing samples in this assay were found to be sample 2 and sample 3 for limiting both slime and the overall mass increase on the surface. The level of mass adsorption for the anti-fouling paint and the control group infers that additional mass dramatically increased during the 7 day period. The increase in mass could be attributed to the process of sediment attachment also adhering to the substrates. Similarly, Needoba *et al.* experienced an increase in the mass on dissolved oxygen sensors³⁰ with a material that is transparent and also semi-permeable. However, in Fig. 7 slime is the observed contributor to the effect of mass as we have quantified and corroborated both assays simultaneously representing a straightforward and accurate way to show that overall biomass is linked to slime adsorption on the surface. It should be noted that even though sample 1 has a similar topography to sample 2 it exhibited a significant increase in mass in comparison. This effect seemed much greater than the relative differences in absorption for the other assays and may be related to regions on the sample surface where slime nucleation has taken hold and allowed growth from those points. As it was not possible to exclude vein regions of the insect wings in our replication process (and thus these components are also replicated) it is possible that nucleation (in relation to slime) takes place at these locations. The vein regions on the cicada (sample 1) are significantly larger in size compared to the other two samples and this supports this speculation; however,

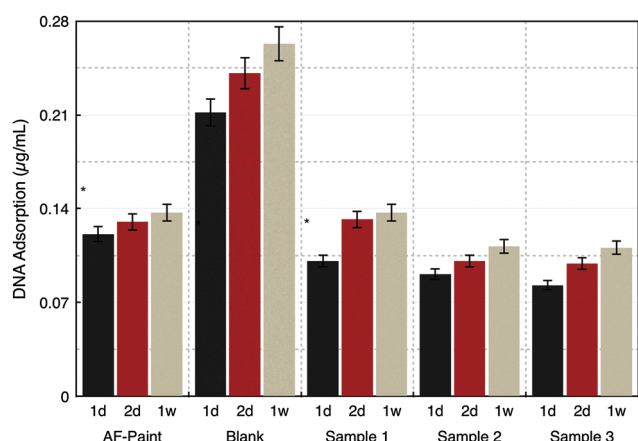


Fig. 7 Chart showing overall DNA adsorption for the materials following the 7 day marine biofouling study. All replicated insect wing samples have resisted DNA adsorption and exhibited equal or superior performance when compared to the AF paint. ($n = 14$, +1 SD).

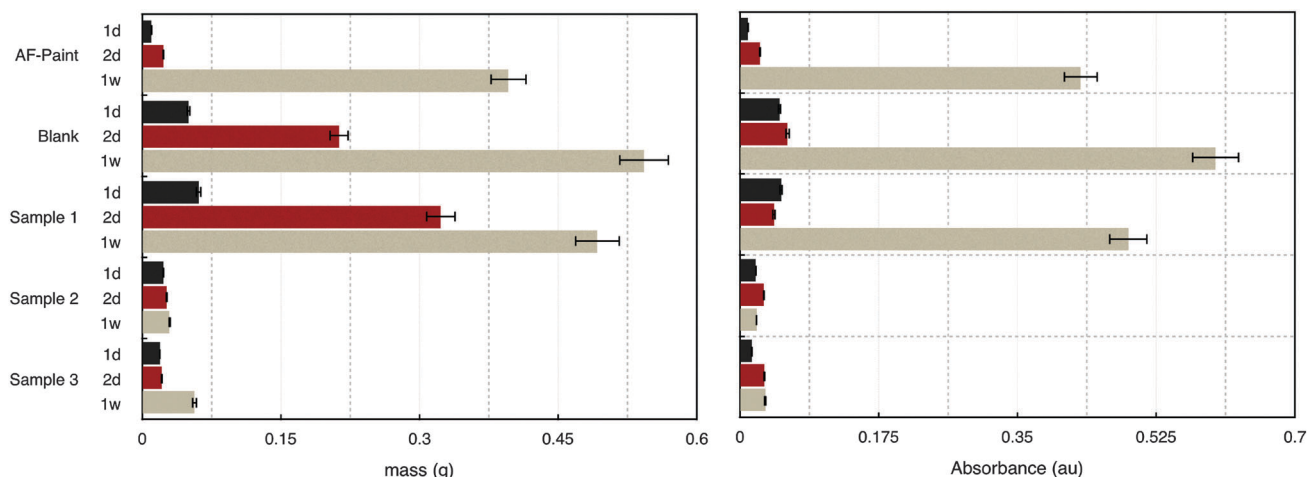


Fig. 8 Combinational bar plots to show the relationship between mass increase (% increase) and overall slime contribution given in absorbance units. ($n = 14$, +1 SD).

further future studies eliminating vein components may be required to ascertain the mechanisms relating to attachment.

We have previously reported the use of the slime and mass assay as an initial screening tool to test antifouling materials we produce.³¹ As a standalone assay, the effectiveness of the slime assay merely remains as a screening tool. When combined, all the quantitative assays clearly show that sample 3 is performing well as an antifouling topographically active surface when compared to the chemically active antifouling paint (Fig. 8).

The difference between samples 1 and 2 for slime and mass increase after 7 days is considerably different. The surface structure of the replicated surfaces is also different in nano-pillar height, demonstrating that perhaps this height difference induces this effect in resisting early stage biofouling.

Conclusions

Antifouling materials have had to undergo a series of legislative changes in direction since 2008. The use of non-toxic and non-biocide release based coatings is appealing to a number of water industries. We have focussed this study on a non-toxic, non-release biomimetic approach to better understand some of the principles of replication of natural surfaces. Our paper has demonstrated the performance of coatings in real seawater environments and is therefore unlike most studies of this type, which rely on pure-culture systems. Indeed, this is the first time that replicated insect wings have been taken through marine biofouling assays as potential candidates for antifouling materials. For each of the artificial insect wings presented in this paper, each has shown improved resistance to biofouling when compared to the epoxy blank control group. In particular, the most disordered topographically structured replicate based on the dragonfly wing exhibited the most efficient form for anti-fouling in almost all cases. For future studies the materials should adopt a two-fold approach incorporating marine secondary metabolites to demonstrate the effectiveness of surface topography and surface chemistry. In addition, the mechanisms responsible for low adhesion demonstrated on our samples require further study, in particular the stability and nature of any air layer formed at the surfaces. Our results indicate that with further work it may be possible to compete with market leading coatings using environmentally more favourable coatings.

References

- J. Chapman, C. Hellio, T. Sullivan, R. Brown, S. Russell, E. Kitteringham, L. Le Nor and F. Regan, *Int. Biodeterior. Biodegrad.*, 2014, **86**, 6.
- J. Chapman, E. Weir and F. Regan, *Colloids Surf., B*, 2010, **78**, 208.
- C. Hellio, D. De La Broise, L. Dufosse, Y. Le Gal and N. Bourgoignon, *Mar. Environ. Res.*, 2001, **52**, 231.
- A. Bergstrand, G. Rahmani-Monfared, Å. Östlund, M. Nydén and K. Holmberg, *J. Biomed. Mater. Res., Part A*, 2009, **88**, 608.
- A. Kerr, M. Cowling, C. Beveridge, M. Smith, A. Parr, R. Head, J. Davenport and T. Hodgkiess, *Environ. Int.*, 1998, **24**, 331.
- J. Chapman and F. Regan, *J. Appl. Biomater. Biomech.*, 2010, **9**, 176.
- D. M. Yebra, S. Kiil and K. Dam-Johansen, *Prog. Org. Coat.*, 2004, **50**, 75.
- M. E. Callow and J. A. Callow, *Biologist*, 2002, **49**, 1.
- C. Hellio, M. Tsoukatou, J.-P. Marechal, N. Aldred, C. Beaupoil, A. S. Clare, C. Vagias and V. Roussis, *Mar. Biotechnol.*, 2005, **7**, 297.
- H. Bi, W. Zhong, S. Meng, J. Kong, P. Yang and B. Liu, *Anal. Chem.*, 2006, **78**, 3399.
- H.-M. Hu, J. A. Watson, B. W. Cribb and G. S. Watson, *Biofouling*, 2011, **27**, 1125.
- J. A. Watson, H.-M. Hu, B. W. Cribb and G. S. Watson, *Anti-wetting on insect cuticle – structuring to minimise adhesion and weight*, InTech Publishing, 2011.
- K. M. Wisdom, J. A. Watson, X. Qu, F. Liu, G. S. Watson and C.-H. Chen, *Proc. Natl. Acad. Sci. U. S. A.*, 2013, **110**, 7992.
- G. S. Watson and J. A. Watson, *Appl. Surf. Sci.*, 2004, **235**, 139.
- U. Kharchenko, I. Beleneva and E. Dmitrieva, *Int. Biodeterior. Biodegrad.*, 2012, **75**, 68.
- J. Chapman and F. Regan, *Adv. Eng. Mater.*, 2012, **14**, B175.
- Z. Wang, Z. Liang, B. Wang, C. Zhang and L. Kramer, *Composites, Part A*, 2004, **35**, 1225.
- F. Song, K. Xiao, K. Bai and Y. Bai, *J. Mater. Sci. Eng. A*, 2007, **457**, 254.
- L. T. Joshi, D. S. Phillips, C. F. Williams, A. Alyousef and L. Baillie, *Appl. Environ. Microbiol.*, 2012, **78**, 7671.
- M. J. Tobin, L. Puskar, S. H. Nguyen, J. Hasan, H. K. Webb, C. J. Hirschmugl, M. J. Nasse, G. Gervinskas, S. Juodkazis and G. S. Watson, *Spectrosc. Eur.*, 2015, **27**, 15.
- C. J. Weinman, N. Gunari, S. Krishnan, R. Dong, M. Y. Paik, K. E. Sohn, G. C. Walker, E. J. Kramer, D. A. Fischer and C. K. Ober, *Soft Matter*, 2010, **6**, 3237.
- D. Gao, Y. Fu and N. Ren, *Water Res.*, 2013, **47**, 6680.
- M. Sun, A. Liang, G. S. Watson, J. A. Watson, Y. Zheng, J. Ju and L. Jiang, *PLoS One*, 2012, **7**, e35056.
- M. Sun, G. S. Watson, Y. Zheng, J. A. Watson and A. Liang, *J. Exp. Biol.*, 2009, **212**, 3148.
- H.-C. Flemming and J. Wingender, *Nat. Rev. Microbiol.*, 2010, **8**, 623.
- J. Chapman, L. Le Nor, R. Brown, E. Kitteringham, S. Russell, T. Sullivan and F. Regan, *J. Mater. Chem. B*, 2013, **1**, 6194.
- B. Liu and G. C. Bazan, *Chem. Mater.*, 2004, **16**, 4467.
- M. Nosonovsky and B. Bhushan, *Curr. Opin. Colloid Interface Sci.*, 2009, **14**, 270.
- C. L. Tsai, D. J. Schurman and R. L. Smith, *J. Orthop. Res.*, 1988, **6**, 666.
- J. A. Needoba, T. D. Peterson and K. S. Johnson, *Molecular Biological Technologies for Ocean Sensing*, Springer, 2012, p. 73.
- J. Chapman, A. Lawlor, E. Weir, B. Quilty and F. Regan, *J. Membr. Sci.*, 2010, **365**, 180.